

Experiment 36: Foods and fuels

Notes

Most of the energy our body needs comes from fats and carbohydrates. Carbohydrates are broken down in the intestines to glucose. The glucose is transported in the blood to cells where it is oxidised to produce CO_2 , H_2O and energy:



The breakdown of fats also produces CO_2 and H_2O .

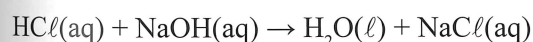
Any excess sugar in the body is stored as fats. About 100 kJ per kilogram of body weight per day is required to keep the body functioning at a minimum level.

Compound	Fuel Value (kJ g^{-1})
Fats	38
Carbohydrates	17
Proteins	17

Most chemical reactions used for the production of heat are combustion reactions. The energy released when 1g of material is burnt is its fuel value. Since all heats of combustion are exothermic, fuel values are reported as positive.

Not all chemical reactions are combustion reactions. Yet every chemical reaction has a particular heat of reaction. This is usually expressed as the molar heat of reaction. The molar heat of oxidation of glucose shown earlier is 2 816 kJ. How are these values determined?

Consider the neutralisation reaction between hydrochloric acid and sodium hydroxide solution:



The molar heat of reaction for this reaction is the heat released when one mole of hydrochloric acid is neutralised with one mole of sodium hydroxide solution.

In this activity you will experimentally determine the molar heat of this neutralisation reaction by measuring the temperature increase that occurs when solutions of hydrochloric acid and sodium hydroxide are mixed. This experiment is outside the requirements of stage 2 units.

Equipment

polystyrene cup
graduated cylinder (100 mL)
thermometer (-10 to 110°C)
hydrochloric acid $[\text{HCl}]$ 1 mol L^{-1} (50 mL)
sodium hydroxide solution $[\text{NaOH}]$ 1 mol L^{-1} (50 mL)